

Lie Theory: Chapter 2 Homework

Connor Mooneyhan

1. Let I and J be ideals of a Lie algebra L . Prove that $I \cap J$ is a subspace of L .

Proof. Given $a, b \in F$ and $x, y \in I \cap J$, we want to show that $ax + by \in I \cap J$. Indeed, $ax + by \in I$ since I is a subspace by virtue of being an ideal, and $ax + by \in J$ for the same reason, so $ax + by \in I \cap J$. $\square$

2. Prove the Isomorphism Theorem (part a).

Proof. Given Lie algebras L_1 and L_2 together with a homomorphism $f : L_1 \rightarrow L_2$, define $g : L_1 / \ker f \rightarrow L_2$ by $g(x + \ker f) = f(x)$. We first show that this is well-defined.

If $x + \ker f = y + \ker f$, we have $y - x \in \ker f$, so

$$\begin{aligned} g(x + \ker f) &= f(x) \\ &= f(x) + 0 \\ &= f(x) + f(y - x) \\ &= f(x + y - x) \\ &= f(y) \\ &= g(y + \ker f), \end{aligned}$$

so g is well-defined.

We now show that it is a linear map by taking, for any $x, y \in L_1$,

$$\begin{aligned} g(a(x + \ker f) + b(y + \ker f)) &= g((ax + \ker f) + (by + \ker f)) \\ &= g((ax + by) + \ker f) \\ &= f(ax + by) \\ &= af(x) + bf(y) \\ &= ag(x + \ker f) + bg(y + \ker f). \end{aligned}$$

It is furthermore a Lie algebra homomorphism because

$$\begin{aligned} g([x + \ker f, y + \ker f]) &= g([x, y] + \ker f) \\ &= f([x, y]) \\ &= [f(x), f(y)] \\ &= [g(x + \ker f), g(y + \ker f)]. \end{aligned}$$

Finally, we show bijectivity onto its image. It is surjective onto $\text{im } g = \text{im } f$ by definition. For injectivity, let $x, y \in L_1$ such that $g(x + \ker f) = g(y + \ker f)$. Then

$$\begin{aligned} 0 &= g(x + \ker f) - g(y + \ker f) \\ &= f(x) - f(y) \\ &= f(x - y), \end{aligned}$$

so $x - y \in \ker f$, from which it follows that $x + \ker f = y + \ker f$. Thus f is a bijective Lie algebra homomorphism $L_1 / \ker f \rightarrow \text{im } f$, i.e. an isomorphism. $\square$

3. Prove the statements on page 14 (just above example 2.4):

- (a) Prove that J (as defined on page 14) is an ideal of L .
- (b) Prove that J contains I .

Proof. Let K be an ideal of L/I and define $J := \{z \in L : z + I \in K\}$.

- (a) Given $a, b \in F$ and $x, y \in J$,

$$\begin{aligned} (ax + by) + I &= (ax + I) + (by + I) \\ &= a(x + I) + b(y + I), \end{aligned}$$

so since $x + I$ and $y + I$ are in K , this is a linear combination of elements of K , which is itself in K as desired. Thus J is a linear subspace.

Now if $x \in J$ and $y \in L$, we have

$$[x, y] + I = [x + I, y + I],$$

so since $x + I \in K$ and K is an ideal, we get $[x, y] + I \in K$ and thus $[x, y] \in J$.

- (b) Given $x \in I$, $x + I = I$, so since I is the identity in L/I and every ideal contains the identity, $x + I \in K$ and thus $x \in J$.

$\square$

4. Suppose L_1 and L_2 are Lie algebras. Let $L := \{(x_1, x_2) : x_i \in L_i\}$ be the direct sum of their underlying vector spaces.

- (a) Show that if we define

$$[(x_1, x_2), (y_1, y_2)] := ([x_1, y_1], [x_2, y_2])$$

then L becomes a Lie algebra, the *direct sum* of L_1 and L_2 . We denote this by $L = L_1 \oplus L_2$.

- (b) Show that if $L = L_1 \oplus L_2$ then $Z(L) = Z(L_1) \oplus Z(L_2)$ and $L' = L'_1 \oplus L'_2$.

Proof. (a) Since we already know that L is a vector space, we will show that the bracket as defined is a valid Lie bracket. Given $(x_1, x_2) \in L$, we have

$$\begin{aligned} [(x_1, x_2), (x_1, x_2)] &= ([x_1, x_1], [x_2, x_2]) \\ &= (0, 0) \end{aligned}$$

as desired. For (y_1, y_2) and (z_1, z_2) also in L , we have

$$\begin{aligned} &[(x_1, x_2), [(y_1, y_2), (z_1, z_2)]] + [(y_1, y_2), [(z_1, z_2), (x_1, x_2)]] + [(z_1, z_2), [(x_1, x_2), (y_1, y_2)]] \\ &= [(x_1, x_2), ([y_1, z_1], [y_2, z_2])] + [(y_1, y_2), ([z_1, x_1], [z_2, x_2])] + [(z_1, z_2), ([x_1, y_1], [x_2, y_2])] \\ &= ([x_1, [y_1, z_1]], [x_2, [y_2, z_2]]) + ([y_1, [z_1, x_1]], [y_2, [z_2, x_2]]) + ([z_1, [x_1, y_1]], [z_2, [x_2, y_2]]) \\ &= ([x_1, [y_1, z_1]] + [y_1, [z_1, x_1]] + [z_1, [x_1, y_1]], [x_2, [y_2, z_2]] + [y_2, [z_2, x_2]] + [z_2, [x_2, y_2]]) \\ &= (0, 0) \end{aligned}$$

as desired.

(b) If $(x_1, x_2) \in Z(L)$, then given any $(y_1, y_2) \in L$,

$$(0, 0) = [(x_1, x_2), (y_1, y_2)] = ([x_1, y_1], [x_2, y_2]),$$

so $[x_1, y_1] = [x_2, y_2] = 0$, and since y_1 and y_2 were arbitrary, this means $(x_1, x_2) \in Z(L_1) \oplus Z(L_2)$. Also, if $(x_1, x_2) \in Z(L_1) \oplus Z(L_2)$, then for any $(y_1, y_2) \in L$,

$$[(x_1, x_2), (y_1, y_2)] = ([x_1, y_1], [x_2, y_2]) = (0, 0),$$

so $(x_1, x_2) \in Z(L)$. Thus by mutual inclusion, $Z(L) = Z(L_1) \oplus Z(L_2)$.

Now if $x \in L'$, then $x = \sum_{i < j} a_{ij}[x^i, x^j]$ for some $\{x^1, x^2, \dots, x^n\} \subset L$ where the superscripts are to be interpreted just as indexing notation, not exponents, and where each $a_{ij} \in F$. We have,

$$\begin{aligned} x &= \sum_{i < j} a_{ij}[x^i, x^j] \\ &= \sum_{i < j} a_{ij}[(x_1^i, x_2^i), (x_1^j, x_2^j)] \\ &= \sum_{i < j} a_{ij}([x_1^i, x_1^j], [x_2^i, x_2^j]) \\ &= \sum_{i < j} (a_{ij}[x_1^i, x_1^j], a_{ij}[x_2^i, x_2^j]) \\ &= \left(\sum_{i < j} a_{ij}[x_1^i, x_1^j], \sum_{i < j} a_{ij}[x_2^i, x_2^j] \right) \in L'_1 \oplus L'_2. \end{aligned}$$

Running that argument backwards, starting with an element of $L'_1 \oplus L'_2$, shows the reverse containment, so we have $L' = L'_1 \oplus L'_2$. □

5. For each pair of the following Lie algebras over $\mathbb{R}$, decide whether or not they are isomorphic:

- (a) the Lie algebra $\mathbb{R}_\wedge^3$ with Lie bracket given by the vector product;
- (b) the upper triangular 2×2 matrices over $\mathbb{R}$;
- (c) the strictly upper triangular 3×3 matrices over $\mathbb{R}$;
- (d) $L = \{x \in \mathfrak{gl}(3, \mathbb{R}) : x^t = -x\}$.

Proof. We note a few things up front. All four candidates have dimension 3. We now look at the dimension of the derived algebra for each.

L	$\dim(L')$
$\mathbb{R}_\wedge^3$	3; every vector in $\mathbb{R}^3$ can be expressed as the product of two others.
$b(2, \mathbb{R})$	1; the bracket of two upper triangular matrices is <i>strictly</i> upper triangular, as we have seen, leaving us with nonzero entries in the top-right entry only.
$n(3, \mathbb{R})$	1; similar to the above case, where the bracket leaves us with nonzero entries in the top-right entry only.
$\{x^t = -x\}$	3; the matrices here look like $\begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}$, so there are three basis elements, each taking $a = 1$, $b = 1$, and $c = 1$ respectively, with zeroes everywhere else.

Thus (a) and (d) are isomorphic to each other since all 3-dimensional Lie algebras L with $L = L'$ are isomorphic.

A general element of $b(2, \mathbb{R})$ is a strictly upper triangular 2×2 matrix $A = \begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix}$, which has determinant 0, so when we bracket it with a general element $B = \begin{pmatrix} x & y \\ 0 & z \end{pmatrix} \in b(2, \mathbb{R})$, we get

$$\begin{aligned}
[A, B] &= \begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x & y \\ 0 & z \end{pmatrix} - \begin{pmatrix} x & y \\ 0 & z \end{pmatrix} \begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & az \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & ax \\ 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & a(z - x) \\ 0 & 0 \end{pmatrix},
\end{aligned}$$

which can be nonzero (say, if $a = 1, z = 2, x = 1$), so $b(2, \mathbb{R})' \not\subset Z(b(2, \mathbb{R}))$.

Meanwhile, a general element $A = \begin{pmatrix} 0 & 0 & a \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ of $n(3, \mathbb{R})'$ bracketed with a general element

$B = \begin{pmatrix} 0 & x & y \\ 0 & 0 & z \\ 0 & 0 & 0 \end{pmatrix}$ of $n(3, \mathbb{R})$ yields

$$\begin{aligned} [A, B] &= \begin{pmatrix} 0 & 0 & a \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & x & y \\ 0 & 0 & z \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & x & y \\ 0 & 0 & z \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & a \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \end{aligned}$$

so $n(3, \mathbb{R})' \subset Z(n(3, \mathbb{R}))$. Thus (b) and (c) are not isomorphic. $\square$

6. Let A and B be $n \times n$ matrices with complex entries.

- (a) Is it true that $\text{tr}(AB) = \text{tr}(A)\text{tr}(B)$? If true, prove it. If false, provide a counterexample.
- (b) Is it true that $\text{tr}(AB) = \text{tr}(BA)$? If true, prove it. If false, provide a counterexample.
- (c) Use your answers to (a) and/or (b) to show that if $x \in \mathfrak{gl}_S(n, \mathbb{C})$ for some $n \times n$ invertible matrix S , then $\text{tr}(x) = 0$.

Proof.

(a) No. Consider

$$\text{tr} \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) = \text{tr} \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) = 2$$

and

$$\text{tr} \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) \text{tr} \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) = 2 \cdot 2 = 4.$$

(b) Yes.

$$\begin{aligned} \text{tr}(AB) &= \sum_{i=1}^n (AB)_{ii} \\ &= \sum_{i=1}^n \sum_{j=1}^n A_{ij} B_{ji} \\ &= \sum_{i=1}^n \sum_{j=1}^n A_{ji} B_{ij} \\ &= \sum_{i=1}^n \sum_{j=1}^n B_{ij} A_{ji} \\ &= \sum_{i=1}^n (BA)_{ii} \\ &= \text{tr}(BA), \end{aligned}$$

where the third equality comes not from term-by-term equality as ordered, but from realizing that since we are summing over all i and j between 1 and n inclusive, each term from the sum is represented once in both versions, so we can use commutativity to reorder the sum conveniently to fit our needs.

- (c) If $x \in \mathfrak{gl}_S(n, \mathbb{C})$, then $x^T S = -Sx$. We will also use the fact that $\text{tr}(y) = \text{tr}(y^T)$ for any matrix y since the diagonal doesn't change when you take the transpose. We have

$$\begin{aligned} \text{tr}(x) &= \text{tr}(S^{-1}Sx) \\ &= -\text{tr}(S^{-1}(x^T S)) \\ &= -\text{tr}(x^T S S^{-1}) \\ &= -\text{tr}(x^T) \\ &= -\text{tr}(x), \end{aligned}$$

so $\text{tr}(x) = 0$.

□

7. Let I be an ideal of a Lie algebra L . Let B be the centralizer of I in L ; that is

$$B = C_L(I) = \{x \in L : [x, a] = 0 \text{ for all } a \in I\}.$$

Show that B is an ideal of L .

Proof. First, we show closure under linear combinations. If $a, b \in F$, $x, y \in B$, and $z \in I$,

$$\begin{aligned} [ax + by, z] &= a[x, z] + b[y, z] \\ &= 0 + 0 \\ &= 0, \end{aligned}$$

so $ax + by \in B$.

Now if $x \in B$ and $y \in L$, then given any $z \in I$

$$\begin{aligned} [[x, y], z] &= -[z, [x, y]] \\ &= [x, [y, z]] + [y, [z, x]]. \end{aligned}$$

Note that $[y, z] \in I$ since $z \in I$, so $[x, [y, z]] = 0$, and $[z, x] = 0$ since $x \in B$ so $[y, [z, x]] = [y, 0] = 0$, so we indeed have $[[x, y], z] = 0$, meaning $[x, y] \in B$. Thus B is an ideal. □

8. Prove the Isomorphism Theorem (part b).

Proof. Define $f : I + J \rightarrow \frac{I}{I \cap J}$ by $i + j \mapsto i + I \cap J$. This is well defined because if $i_1 + j_1 = i_2 + j_2$ then $i_2 - i_1 = j_1 - j_2 \in I \cap J$, so

$$\begin{aligned} f(i_1 + j_1) &= i_1 + I \cap J \\ &= i_1 + (i_2 - i_1 + I \cap J) \\ &= (i_1 + i_2 - i_1) + I \cap J \\ &= i_2 + I \cap J \\ &= f(i_2 + j_2). \end{aligned}$$

It is linear because

$$\begin{aligned}
f(a(i_1 + j_1) + b(i_2 + j_2)) &= f(ai_1 + bi_2 + aj_1 + bj_2) \\
&= ai_1 + bi_2 + I \cap J \\
&= (ai_1 + I \cap J) + (bi_2 + I \cap J) \\
&= a(i_1 + I \cap J) + b(i_2 + I \cap J) \\
&= af(i_1 + j_1) + bf(i_2 + j_2),
\end{aligned}$$

and

$$\begin{aligned}
f([i_1 + j_1, i_2 + j_2]) &= f([i_1, i_2] + [i_1, j_2] + [j_1, i_2] + [j_1, j_2]) \\
&= [i_1, i_2] + I \cap J \\
&= [i_1 + I \cap J, i_2 + I \cap J] \\
&= [f(i_1 + j_1), f(i_2 + j_2)],
\end{aligned}$$

where the second equality results from the fact that the final three terms of the argument of the RHS of the first equality are in J by virtue of J being an ideal. Thus f is a Lie algebra homomorphism.

We will now show that $\ker f = J$. Indeed, if $i \in I$ is not in J , then i is not in $I \cap J$, so $f(i + j) \neq I \cap J$, i.e. it is nonzero, for any choice of $j \in J$. If $i \in I$ is in J , then $i \in I \cap J$, so $f(i + j) = i + I \cap J = I \cap J$, which is zero in $\frac{I}{I \cap J}$. Clearly if $i + j$ is of the form $0 + j$, then $f(0 + j) = 0 + I \cap J = I \cap J$. Thus $\ker f = J$.

We also see immediately that f is surjective, since given $i + I \cap J \in \frac{I}{I \cap J}$, we have $f(i) = i + I \cap J$. Thus the first isomorphism theorem gives us that

$$\frac{I + J}{J} = \frac{I + J}{\ker f} \cong \text{im } f = \frac{I}{I \cap J}.$$

□

9. Prove the Isomorphism Theorem (part c).

Proof. Let I and J be ideals of a Lie algebra L such that $I \subset J$.

We first show that J/I is an ideal of L/I . Indeed, given $j + I \in J/I$ and $\ell + I \in L/I$, we have

$$[j + I, \ell + I] = [j, \ell] + I \in J/I$$

since $[j, \ell] \in J$ because J is an ideal.

Now we define $f : L/I \rightarrow L/J$ by $\ell + I \mapsto \ell + J$, and we want to show that this is surjective homomorphism and that its kernel is J/I . First we show that f is well-defined: if $\ell_1 + I = \ell_2 + I$ then $\ell_2 - \ell_1 \in I \subset J$, so

$$\begin{aligned}
f(\ell_1 + I) &= \ell_1 + J \\
&= \ell_1 + (\ell_2 - \ell_1 + J) \\
&= (\ell_1 + \ell_2 - \ell_1) + J \\
&= \ell_2 + J \\
&= f(\ell_2 + I).
\end{aligned}$$

Now we show that it is linear:

$$\begin{aligned}
f(a(\ell_1 + I) + b(\ell_2 + I)) &= f((a\ell_1 + b\ell_2) + I) \\
&= (a\ell_1 + b\ell_2) + J \\
&= (a\ell_1 + J) + (b\ell_2 + J) \\
&= a(\ell_1 + J) + b(\ell_2 + J) \\
&= af(\ell_1 + I) + bf(\ell_2 + I).
\end{aligned}$$

Then we show that it preserves brackets:

$$\begin{aligned}
f([\ell_1 + I, \ell_2 + I]) &= f([\ell_1, \ell_2] + I) \\
&= [\ell_1, \ell_2] + J \\
&= [\ell_1 + J, \ell_2 + J] \\
&= [f(\ell_1 + I), f(\ell_2 + I)].
\end{aligned}$$

So f is a Lie algebra homomorphism.

Now, given any $\ell + J \in L/J$, $f(\ell + I) = \ell + J$, so we have surjectivity. Finally, if $f(\ell + I) = \ell + J = J$, then we must have $\ell \in J$, so $\ker f \subset J/I$. On the other hand, if $j \in J$, then $f(j + I) = j + J = J$, so we have $\ker f = J/I$ as desired.

Thus the first isomorphism theorem gives us that

$$\frac{L/I}{J/I} = \frac{L/I}{\ker f} \cong \text{im } f = \frac{L}{J}.$$

□

10. Let F be a field. Show that the derived algebra of $\mathfrak{gl}(n, F)$ is $\mathfrak{sl}(n, F)$.

Proof. Let $A, B \in \mathfrak{gl}(n, F)$. Since $\text{tr}(AB) = \text{tr}(BA)$ and trace is linear,

$$\begin{aligned}
\text{tr}([A, B]) &= \text{tr}(AB - BA) \\
&= \text{tr}(AB) - \text{tr}(BA) \\
&= \text{tr}(AB) - \text{tr}(AB) \\
&= 0.
\end{aligned}$$

Thus $[A, B] \in \mathfrak{sl}(n, F)$ for all $A, B \in \mathfrak{gl}(n, F)$, so their span is contained in the subspace $\mathfrak{sl}(n, F)$, i.e. $\mathfrak{gl}(n, F)' \subset \mathfrak{sl}(n, F)$.

To show the other direction, we show that the standard basis elements of $\mathfrak{sl}(n, F)$ are contained in $\mathfrak{gl}(n, F)'$. We know that for a, b, c, d between 1 and n , $e_{ab}e_{cd} = \delta_{bc}e_{ad}$ since all entries are zero except perhaps entry ad , which is nonzero if and only if $b = c$. Given this, we can express the basis elements of $\mathfrak{sl}(n, F)$ as

$$e_{ij} = e_{ii}e_{ij} = e_{ii}e_{ij} - e_{ij}e_{ii} = [e_{ii}, e_{ij}] \in \mathfrak{gl}(n, F)'$$

when $i \neq j$ and

$$e_{ii} - e_{i+1, i+1} = e_{i, i+1}e_{i+1, i} - e_{i+1, i}e_{i, i+1} = [e_{i, i+1}, e_{i+1, i}] \in \mathfrak{gl}(n, F)'.$$

Thus $\mathfrak{sl}(n, F) = \mathfrak{gl}(n, F)'$ by mutual containment.

□